



MAT-121

Differential Calculus

By Das and Mukherjee

51st Edition

Chapter 02: Continuity

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Continuity

❖ Definition:

- ▶ A function $f(x)$ is said to be continuous for $x = a$, provided $\lim_{x \rightarrow a} f(x)$ exists, is finite and is equal to $f(a)$.
- ▶ In other words, for $f(x)$ to be continuous at $x = a$,
$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a) .$$
- A function $f(x)$ is not continuous at a point is said to have a discontinuous at that point.

Example-1 A function $f(x)$ is define as follows:

$$f(x) = \begin{cases} \frac{1}{2} - x, & \text{when } 0 < x < \frac{1}{2} \\ \frac{1}{2}, & \text{for } x = \frac{1}{2} \\ \frac{3}{2} - x, & \text{for } \frac{1}{2} < x < 1 \end{cases} ; \text{ show that } f(x) \text{ is discontinuous at } x = \frac{1}{2}.$$

Solution:

$$\text{We have } f(x) = \begin{cases} \frac{1}{2} - x, & \text{when } 0 < x < \frac{1}{2} \\ \frac{1}{2}, & \text{for } x = \frac{1}{2} \\ \frac{3}{2} - x, & \text{for } \frac{1}{2} < x < 1 \end{cases}$$

- When $x = \frac{1}{2}$.

$$\text{Functional Value F.V} = f(x) = \frac{1}{2}$$

$$\text{So, } F.V = f\left(\frac{1}{2}\right) = \frac{1}{2}$$

Example-1 Cont.

$$\begin{aligned} \text{▶ L.H.L} &= \lim_{x \rightarrow \left(\frac{1}{2}\right)^-} f(x) = \lim_{x \rightarrow \left(\frac{1}{2}\right)^-} \left(\frac{1}{2} - x \right) = \frac{1}{2} - \frac{1}{2} = 0 \\ \text{▶ R.H.L} &= \lim_{x \rightarrow \left(\frac{1}{2}\right)^+} f(x) = \lim_{x \rightarrow \left(\frac{1}{2}\right)^+} \left(\frac{3}{2} - x \right) = \frac{3}{2} - \frac{1}{2} = 1 \end{aligned}$$

Since $\text{L.H.L} \neq \text{R.H.L} \neq \text{F.V}$

Hence $f(x)$ is discontinuous at $x = \frac{1}{2}$. (Showed)

Example-2 If $f(x) = \begin{cases} x^2, & \text{when } 0 < x < 1 \\ x, & \text{when } 1 \leq x < 2 \\ \frac{x^3}{4}, & \text{when } 2 \leq x < 3 \end{cases}$; show that $f(x)$ is continuous

at $x = 1$ and $x = 2$.

Solution:

We have $f(x) = \begin{cases} x^2, & \text{when } 0 < x < 1 \\ x, & \text{when } 1 \leq x < 2 \\ \frac{x^3}{4}, & \text{when } 2 \leq x < 3 \end{cases}$;

- When $x = 1$

Functional Value F.V = $f(x) = x$

So, F.V = $f(1) = 1$

$$\text{L.H.L} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1^2 = 1$$

$$\text{R.H.L} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x = 1$$

Since $\text{L.H.L} = \text{R.H.L} = \text{F.V}$

So, $f(x)$ is continuous at $x = 1$.

Example-2 Cont.

- Again when $x = 2$

$$\text{Functional Value F.V} = f(x) = \frac{x^3}{4}$$

$$\text{Thus, F.V} = f(2) = \frac{2^3}{4} = \frac{8}{4} = 2$$

$$\text{L.H.L} = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \left(\frac{x^3}{4} \right) = 2$$

$$\text{R.H.L} = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \left(\frac{x^3}{4} \right) = \frac{2^3}{4} = \frac{8}{4} = 2$$

Since L.H.L = R.H.L = F. V

So, $f(x)$ is continuous at $x = 2$.

Hence Shown

Ex.3 A function $\varphi(x)$ is defined as follows: $\varphi = \begin{cases} x^2, & \text{for } x < 1 \\ 2.5, & \text{for } x = 1 \\ x^2 + 2, & \text{for } x > 1 \end{cases}$;

is $\varphi(x)$ continuous at $x = 1$?

Solution:

We have $\varphi(x) = \begin{cases} x^2, & \text{for } x < 1 \\ 2.5, & \text{for } x = 1 \\ x^2 + 2, & \text{for } x > 1 \end{cases}$

- When $x = 1$.

Functional Value F.V = $\varphi(x) = 2.5$

So, F.V = $\varphi(1) = 2.5$

L.H.L = $\lim_{x \rightarrow 1^-} \varphi(x) = \lim_{x \rightarrow 1^-} x^2 = 1^2 = 1$

R.H.L = $\lim_{x \rightarrow 1^+} \varphi(x) = \lim_{x \rightarrow 1^+} (x^2 + 2) = 1^2 + 2 = 3$

Since $L.H.L \neq R.H.L \neq F.V$

So, $\varphi(x)$ does not continuous at $x = 1$.

Ex.4 A function $f(x)$ is defined as follows: $f(x) = \begin{cases} -x, & \text{when } x \leq 0 \\ x, & \text{when } 0 < x < 1; \\ 2 - x, & \text{when } x \geq 1 \end{cases}$

Show that $f(x)$ is continuous at $x = 0$ and $x = 1$.

Solution: We have $f(x) = \begin{cases} -x, & \text{when } x \leq 0 \\ x, & \text{when } 0 < x < 1 \\ 2 - x, & \text{when } x \geq 1 \end{cases}$

- When $x = 0$

Functional Value F.V = $f(x) = -x$

So, F.V = $f(0) = -0 = 0$

▶ L.H.L = $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x) = 0$

▶ R.H.L = $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuity at $x = 0$.

Ex.4 Cont.

- Again when $x = 1$

Functional Value F.V = $f(x) = 2 - x$

So, F.V = $f(1) = 2 - 1 = 1$

- ▶ L.H.L = $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x = 1$
- ▶ R.H.L = $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2 - x) = 2 - 1 = 1$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuity at $x = 1$.

Hence showed

Ex.5 A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 3 + 2x, & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x, & \text{for } 0 \leq x < \frac{3}{2} \\ -3 + 2x, & \text{for } x \geq \frac{3}{2} \end{cases} ; \text{ Show that } f(x) \text{ is continuous at } x = 0 \text{ and discontinuous at } x = \frac{3}{2}.$$

Solution: Given $f(x) = \begin{cases} 3 + 2x, & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x, & \text{for } 0 \leq x < \frac{3}{2} \\ -3 + 2x, & \text{for } x \geq \frac{3}{2} \end{cases}$

- When $x = 0$

Functional Value F.V = $f(x) = 3 - 2x$

So, F.V = $f(0) = 3 - 2 \times 0 = 3$

L.H.L = $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (3 + 2x) = 3 + 2 \times 0 = 3$

R.H.L = $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (3 - 2x) = 3 - 2 \times 0 = 3$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuity at $x = 0$.

Ex.5 Cont.

- Again when $x = \frac{3}{2}$

$$\text{Functional Value F.V} = f(x) = -3 - 2x$$

$$\text{So, F.V} = f\left(\frac{3}{2}\right) = -3 - 2 \times \frac{3}{2} = -3 - 3 = -6$$

$$\text{L.H.L} = \lim_{x \rightarrow \left(\frac{3}{2}\right)^-} f(x) = \lim_{x \rightarrow \left(\frac{3}{2}\right)^-} (3 - 2x) = 3 - 2 \times \frac{3}{2} = 3 - 3 = 0$$

$$\text{R.H.L} = \lim_{x \rightarrow \left(\frac{3}{2}\right)^+} f(x) = \lim_{x \rightarrow \left(\frac{3}{2}\right)^+} (-3 - 2x) = -3 - 2 \times \frac{3}{2} = -3 - 3 = -6$$

$$\text{Since L.H.L} \neq \text{R.H.L} = \text{F.V}$$

So the function $f(x)$ is discontinuity at $x = \frac{3}{2}$.

Ex.6 A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 1, & \text{for } x < 0 \\ 1 + \sin x, & \text{for } 0 < x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & \text{for } x \geq \frac{\pi}{2} \end{cases} ; \text{ Show that } f(x) \text{ is continuous at } x = \frac{\pi}{2}.$$

► **Solution:** We have $f(x) = \begin{cases} 1, & \text{for } x < 0 \\ 1 + \sin x, & \text{for } 0 < x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & \text{for } x \geq \frac{\pi}{2} \end{cases} ;$

► When $x = \frac{\pi}{2}$

$$\text{Functional Value F.V} = f(x) = 2 + \left(x - \frac{\pi}{2}\right)^2$$

$$\text{So, F.V} = f\left(\frac{\pi}{2}\right) = 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2 = 2$$

$$\begin{aligned}\text{L.H.L} &= \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} f(x) = \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} (1 + \sin x) \\ &= 1 + \sin \frac{\pi}{2} = 1 + 1 = 2\end{aligned}$$

$$\text{R.H.L} = \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^+} f(x) = \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^+} \left\{ 2 + \left(x - \frac{\pi}{2}\right)^2 \right\} = 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2 = 2$$

Since L.H.L = R.H.L = F.V

So the function $f(x)$ is continuity at $x = \frac{\pi}{2}$.

(Hence Showed)

Ex.7 A function $f(x)$ is defined as follows: $f(x) = x$ when $x > 0$, $f(0) = 0$, $f(x) = -x$, when $x < 0$; Prove that $f(x)$ is continuous at $x = 0$.

Solution: We have $f(x) = \begin{cases} x, & \text{for } x > 0 \\ 0, & \text{for } x = 0 \\ -x, & \text{for } x < 0 \end{cases}$

- When $x = 0$

Functional Value F.V = $f(0) = 0$

$$\text{L.H.L} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x) = 0$$

$$\text{R.H.L} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x) = 0$$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuous at $x = 0$.

(Hence Proved)

Ex.8 A function $f(x)$ is defined as follows: $f(x) = 1, 0$ or -1 according as $x >, =$ or < 0 ; Show that $f(x)$ is discontinuous at $x = 0$.

Solution: We have $f(x) = \begin{cases} 1, & \text{for } x > 0 \\ 0, & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$;

- When $x = 0$

Functional Value F.V = $f(x) = 0$

So, F.V = $f(0) = 0$

$$\text{L.H.L} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-1) = -1$$

$$\text{R.H.L} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (1) = 1$$

Since $\text{L.H.L} \neq \text{R.H.L} \neq \text{F.V}$

Thus the function $f(x)$ discontinuous at $x = 0$.

(Hence Showed)

Ex.9(i) If $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } 1 \leq x \leq 2 \\ x - \frac{x^2}{2} & \text{if } x \geq 2 \end{cases}$; Discuss the continuity at $x = 2$.

Solution: We have $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } 1 \leq x \leq 2 \\ x - \frac{x^2}{2} & \text{if } x \geq 2 \end{cases}$

- When $x = 2$

Functional Value F.V = $f(x) = 2 - x$

So, F.V = $f(2) = 2 - 2 = 0$

▶ L.H.L = $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2 - x) = 2 - 2 = 0$

▶ R.H.L = $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x - \frac{x^2}{2}) = 2 - \frac{2^2}{2} = 2 - 2 = 0$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuous at $x = 2$.

Ex.9(ii) If $f(x) = \begin{cases} \frac{\tan^2 x}{3x} & \text{if } x \neq 0 \\ \frac{2}{3} & \text{if } x = 0 \end{cases}$

Discuss the continuous at $x = 0$.

Solution: Given $f(x) = \begin{cases} \frac{\tan^2 x}{3x} & \text{if } x \neq 0 \\ \frac{2}{3} & \text{if } x = 0 \end{cases}$

- When $x = 0$

Functional Value F.V = $f(x) = \frac{2}{3}$

So, F.V = $f(0) = \frac{2}{3}$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{\tan^2 x}{3x} \right) = \lim_{x \rightarrow 0^-} \left(\frac{\sin^2 x}{\cos^2 x} \times \frac{1}{3x} \right) \\ &= \lim_{x \rightarrow 0^-} \left(\frac{\sin^2 x}{x^2} \times \frac{1}{\cos^2 x} \times \frac{x}{3} \right) \quad \left[\text{Since } \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \right] \end{aligned}$$

Ex.9(ii) Cont.

$$\begin{aligned}\text{L.H.L} &= \lim_{x \rightarrow 0^-} \left(\frac{\sin x}{x} \right)^2 \times \lim_{x \rightarrow 0^-} \left(\frac{1}{\cos^2 x} \right) \times \lim_{x \rightarrow 0^-} \left(\frac{x}{3} \right) \\ &= 1 \times 1^2 \times 0 = 0\end{aligned}$$

$$\text{Similarly, R.H.L} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(\frac{\tan^2 x}{3x} \right) = 0$$

Since L.H.L = R.H.L $\neq$ F.V

Thus $f(x)$ discontinuous at $x = 0$.

Ex.9(iii) If $f(x) = \begin{cases} x^2 + x & \text{for } x < 1 \\ 2 & \text{for } x = 1 \\ 2x^3 - x + 1 & \text{for } 1 < x \leq 2 \end{cases}$; Discuss the continuity at $x = 1$.

Solution: We have $f(x) = \begin{cases} x^2 + x & \text{for } x < 1 \\ 2 & \text{for } x = 1 \\ 2x^3 - x + 1 & \text{for } 1 < x \leq 2 \end{cases}$

- When $x = 1$

Functional Value F.V = $f(x) = 2$

So, F.V = $f(1) = 2$

- ▶ L.H.L = $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + x) = 1^2 + 1 = 2$
- ▶ R.H.L = $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x^3 - x + 1) = 2 \times 1^3 - 1 + 1 = 2$

Since L.H.L = R.H.L = F.V

Thus the function $f(x)$ continuous at $x = 1$.

Ex.9(iv) If $f(x) = \begin{cases} \frac{x^4+4x^3+2x}{\sin x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Discuss the continuity at $x = 0$.

Solution: Given $f(x) = \begin{cases} \frac{x^4+4x^3+2x}{\sin x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

- When $x = 0$

Functional Value F.V = $f(x) = 0$

So, F.V = $f(0) = 0$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{x^4+4x^3+2x}{\sin x} \right) = \lim_{x \rightarrow 0^-} \left\{ \frac{x(x^3+4x^2+2)}{\sin x} \right\} \\ &= \lim_{x \rightarrow 0^-} \left(\frac{(x^3+4x^2+2)}{\frac{\sin x}{x}} \right) \end{aligned}$$

Ex.9(iv) Cont.

$$\begin{aligned} \text{L.H.L} &= \frac{\lim_{x \rightarrow 0^-} (x^3 + 4x^2 + 2)}{\lim_{x \rightarrow 0^-} \left(\frac{\sin x}{x} \right)} & [\text{Since } \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1] \\ &= \frac{0^3 + 4 \times 0^2 + 2}{1} = 2 \end{aligned}$$

$$\text{Similarly, R.H.L} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(\frac{x^4 + 4x^3 + 2x}{\sin x} \right) = 2$$

Since $\text{L.H.L} = \text{R.H.L} \neq \text{F.V}$

Thus $f(x)$ **discontinuous** at $x = 0$.

THANK YOU